

Python Essential Project

A simple, terminal-based arithmetic quiz application developed in Python. Built as a CSE minor project to demonstrate fundamental control flow, input handling, and modular menu-driven execution.

Project Details

- **Branch / Subject:** CSE
- **Academic Session:** 2026–2027
- **Language:** Python 3.x

Overview

The **Python Essential Project** is a lightweight command-line project that tests arithmetic calculations across four basic operations:

1. Addition
2. Subtraction
3. Multiplication
4. Division

The user is prompted for their name, selects a topic from the menu, and answers multiple-choice questions (MCQs). The program checks responses in real time, flags invalid choices, and prints a final score summary.

Features

- **Personalized CLI:** Welcomes the user by name and tracks their score.
- **Menu-Driven:** Clean numeric selection for arithmetic operations.
- **Instant Feedback:** Validates answers immediately (Correct / Wrong / Invalid).
- **Score Accumulator:** Tracks correct attempts and presents the final score at the end.
- **Zero Dependencies:** Runs on standard Python 3 with no external packages needed.

Requirements

- **Python:** Version 3.8 or higher
- **OS:** Windows / Linux / macOS
- **Environment:** Terminal, Command Prompt, or Python IDLE

Project Structure

Python-Essential-Project/

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■■■quiz.py

Python source code

■■■README.md

Project documentation

■■■College_Minor_Project_Report.pdf

Minor project report

How to Run

1. Open your terminal or Command Prompt.

2. Navigate to the folder containing the file:

```
cd path/to/Python-Essential-Project
```

3. Run the script:

```
python quiz.py
```

4. Enter your name, choose a topic (`1-4`), and type `a`, `b`, `c`, or `d` for each question.

Sample Run

Enter your name:

Welcome to the Python Essential Project,

Test your calculation skills!

Choose a topic:

1. Addition

2. Subtraction

3. Multiplication

4. Division

Enter your choice (1-4): 1

here are the addition questions

1: $326 + 457 =$

(a) 773

(b) 783

(c) 793

(d) 763

Enter your answer: b

Correct answer!

2: $648 + 275 =$

(a) 923

(b) 913

(c) 933

(d) 943

Enter your answer: a

Correct answer!

=====

your score is 2

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Thank you for playing the Python Essential Project!

Future Enhancements

- Auto-generate questions dynamically using Python's `random` module.
- Add `try-except` blocks to handle non-numeric inputs during menu selection.
- Save user scores and timestamps into a text or CSV file.